

Lecture 3: Finding Limits Analytically

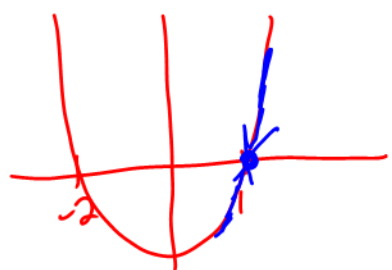
GOAL: Compute limits without guessing.

We start with non-piecewise functions.

To find $\lim_{x \rightarrow a} f(x)$:

Case 1: $f(a)$ is well-defined $\left[\begin{matrix} \text{not undefined} \\ \text{not } \infty, \text{ not } -\infty \end{matrix} \right]$

Ex 1 $\lim_{x \rightarrow 1} (x^2 + x - 2) = 1^2 + 1 - 2 = 0$



In this example, we say $f(x)$ is Continuous at $x=1$. In general,

$$\lim_{x \rightarrow a} f(x) = f\left(\lim_{x \rightarrow a} x\right) = f(a)$$

Case 2: $f(a)$ takes the form $\frac{[\text{Non-Zero}]}{0}$. Then the limit is either ∞ , $-\infty$, or DNE.

Ex 2 $\lim_{x \rightarrow 0} \frac{1}{2x}$

$\lim_{x \rightarrow 0^-} \frac{1}{2x} = -\infty$ (Note: $\frac{1}{2x} < 0$ for $x < 0$)
 $\lim_{x \rightarrow 0^+} \frac{1}{2x} = \infty$ (Note: $\frac{1}{2x} > 0$ for $x > 0$)

	$x < 0$	0	$x > 0$
Top	+		+
Bottom	-		+
Frac	-		+



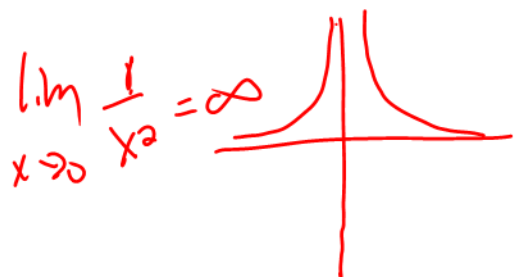
Ex 3 $\lim_{x \rightarrow 0} \frac{1}{x^2}$

$\lim_{x \rightarrow 0^-} \frac{1}{x^2} = \infty$
 $\lim_{x \rightarrow 0^+} \frac{1}{x^2} = \infty$

$\lim_{x \rightarrow -1} \frac{-5}{(x+1)^2} = -\infty$
 $\lim_{x \rightarrow -1^+} \frac{-5}{(x+1)^2} = -\infty$
 $\lim_{x \rightarrow -1^-} \frac{-5}{(x+1)^2} = -\infty$

	$x < 0$	0	$x > 0$
Top	+		+
Bottom	+		+





	$x < -1$	-1	$x > -1$
Top	-		-
Bottom	+		+

Case 3 $f(a)$ gets us an indeterminate form $(\frac{0}{0})$

Ex 4

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1)} = \lim_{x \rightarrow 1} (x^2+x+1) = 1+1+1 = 3$$

Difference of Cubes: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Goal Use Algebra to reduce the limit to Case 1 or case 2.

Ex 5

$$\lim_{x \rightarrow 5} \frac{x^3 - 5x^2}{(x-5)^2} = \lim_{x \rightarrow 5} \frac{x^2(x-5)}{(x-5)^2} = \lim_{x \rightarrow 5} \frac{x^2}{x-5} \text{ DNE}$$

	$x < 5$	5	$x > 5$
Top	+		+
Bottom	-		+

$-\infty$ ∞

$$-0.1 = 4.9 - 5 < 0$$

$$5.1 - 5 = 0.1 > 0$$

Ex 6

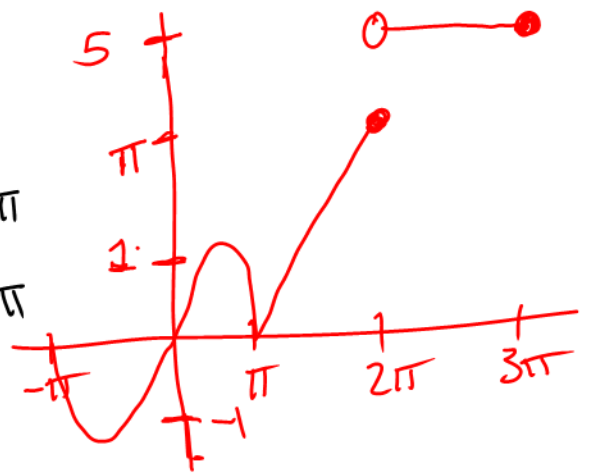
$$\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x^2 + 2x - 8} = \lim_{x \rightarrow 2} \frac{(x-1)(x-2)}{(x+4)(x-2)} = \lim_{x \rightarrow 2} \frac{x-1}{x+4}$$

$$= \frac{2-1}{2+4} = \frac{1}{6}$$

Piecewise Functions

Suppose

$$f(x) = \begin{cases} \sin x & \text{if } -\pi \leq x \leq \pi \\ x - \pi & \text{if } \pi < x \leq 2\pi \\ 5 & \text{if } 2\pi < x \leq 3\pi \end{cases}$$



Case 1 Our point of interest is not a "boundary point"
(bdy. pts in the example are π and 2π).

$$\lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}} \sin x = \sin \frac{\pi}{2} = 1$$

Case 2 The point is a boundary point

$$\begin{aligned} \lim_{x \rightarrow 2\pi} f(x) & \begin{cases} \lim_{x \rightarrow 2\pi^-} f(x) = \lim_{x \rightarrow 2\pi^-} (x - \pi) = 2\pi - \pi = \pi \\ \lim_{x \rightarrow 2\pi^+} f(x) = \lim_{x \rightarrow 2\pi^+} 5 = 5 \end{cases} \end{aligned}$$

$$\lim_{x \rightarrow 2\pi} f(x) \text{ DNE}$$

Limit Laws

Loosely speaking if $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists, then the limits "play well" with arithmetic

$$\bullet \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$\bullet \lim_{x \rightarrow a} [f(x)][g(x)] = \left[\lim_{x \rightarrow a} f(x) \right] \left[\lim_{x \rightarrow a} g(x) \right]$$

$$\bullet \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \leftarrow \neq 0$$

In particular,

$$\bullet \lim_{x \rightarrow a} k \cdot f(x) = k \lim_{x \rightarrow a} f(x) \text{ for constant } k$$

$$\bullet \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \text{ for } n = 1, 2, 3, \dots$$

Ex
If $\lim_{x \rightarrow 3} f(x) = 5$ and $\lim_{x \rightarrow 3} g(x) = 2$, then

$$\begin{aligned} \lim_{x \rightarrow 3} [f(x) + g(x) + 3]^2 &= \left[\lim_{x \rightarrow 3} (f(x) + g(x) + 3) \right]^2 \\ &= \left[\lim_{x \rightarrow 3} f(x) + \lim_{x \rightarrow 3} g(x) + \lim_{x \rightarrow 3} 3 \right]^2 \\ &= [5 + 2 + 3]^2 = 10^2 = 100 \end{aligned}$$

Remark The limits need to exist to apply the Limit Laws.

$$\begin{aligned} \text{Ex/} \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x+x^2} \right) &= \lim_{x \rightarrow 0} \frac{x+x^2-x}{x(x+x^2)} = \lim_{x \rightarrow 0} \frac{x^2}{x(x+x^2)} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{x^2(x+1)} = \lim_{x \rightarrow 0} \frac{1}{x+1} = 1 \end{aligned}$$